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Corporate M&As and Labor Market Concentration: Efficiency Gains or Power Grabs?

Contents

1 Big picture

- **Question.** When mergers increase local labor market concentration, do the gains come from labor market power over workers, or from labor efficiency gains inside the merged firm?
- **Setting.** The paper studies 2,664 completed U.S. public-firm mergers from 1991 to 2016 and measures how much each deal increases concentration in the labor markets where the acquirer and target overlap.
- **Core challenge.** A rise in labor market concentration can be consistent with two very different stories. Under *labor power*, the merged firm suppresses wages and labor demand. Under *labor efficiency*, the merged firm reorganizes workers, reduces redundancies, and improves production.
- **Main empirical innovation.** The paper builds a deal-level measure of merger-induced labor market concentration using establishment-level employment from the National Establishment Time-Series (NETS) database and county-$\times$SIC3 labor markets.
- **Main reduced-form result.** Mergers that increase labor market concentration create more value for the combined firm. For example, over the $(-5, +5)$ window, deals with positive labor market concentration have average combined wealth effects of 2.74% versus 1.50% for deals with no increase in concentration.
- **Channel result.** The overall evidence favors *labor efficiency*, not *labor power*. The relation between labor market concentration and merger gains is stronger when worker skills are more portable, when noncompete agreements are less enforceable, and when union power is weaker. Rival firms exposed in the same labor markets react negatively, while customers and suppliers react positively.
- **Real-effects result.** In overlapping labor markets, merged firms reduce employment and increase IT spending per worker. At the firm level, mergers with larger labor-market overlap are followed by higher revenues, higher operating income, and higher labor productivity, but not by higher operating margins.
- **Economic takeaway.** In this sample of completed U.S. public mergers, increases in labor market concentration appear to generate value mainly because firms use overlapping labor pools to reorganize production more efficiently. That does *not* mean workers are unharmed: some workers in overlapping labor markets are likely displaced.

2 Data and measurement (key objects)

2.1 Sample and data sources

- **Merger sample.** The deal sample comes from SDC Platinum and includes completed mergers announced between 1991 and 2016.
- **Key sample restrictions.**
  - Both acquirer and target must be U.S. public firms covered by CRSP and Compustat.
  - Deals classified as spin-offs, recapitalizations, leveraged buyouts, repurchases, divestitures, self-tenders, acquisitions of assets, acquisitions of remaining interest, or buybacks are excluded.

- A bid is successful if the acquirer owned less than 15% before the contest and at least 50% after it.

- **Core employment data.** The labor-market overlap measure is built from the National Establishment Time-Series (NETS) database, which maps establishments, locations, and employment counts.
- **Financial-market data.** CRSP is used for announcement returns and Compustat for firm characteristics and later real-outcome analysis.
- **Input-output links.** Customer and supplier industries are identified using the BEA input-output tables together with SIC-IO concordances from Fan and Lang.
- **Additional datasets.**
  - Compustat customer segment for "key" customers and suppliers,
  - the Aberdeen Group's Computer Intelligence Technology Database for IT budgets,
  - state-level noncompete enforceability and right-to-work measures,
  - Donangelo's industry mobility measure.

2.2 Defining labor markets and merger overlap

- **Labor market definition.** A labor market is a county-industry cell, where industry is measured at the 3-digit SIC level.
- **Why this matters.** The paper is trying to measure overlap in *hiring markets*, not just in product markets. County-×industry cells are the authors' approximation to local labor markets.
- **Employment shares.** In labor market i, the acquirer and target employment shares are

$$LShare_{A,i} = \frac{Emp_{A,i}}{Emp_i}, \qquad LShare_{T,i} = \frac{Emp_{T,i}}{Emp_i},$$

  where Emp_i is total employment of all firms in that labor market.

2.3 Main concentration measures

The change in labor-market concentration induced by a merger in labor market i is based on the change in HHI:

$$\Delta HHI_i = (LShare_{A,i} + LShare_{T,i})^2 - LShare_{A,i}^2 - LShare_{T,i}^2 = 2LShare_{A,i}LShare_{T,i}.$$

The paper drops the scalar 2 and defines the deal-level measure as

$$\text{Labor Market Concentration} = \sum_{i=1}^{n} \frac{Emp_{A,i} + Emp_{T,i}}{Emp_A + Emp_T} \cdot LShare_{A,i}LShare_{T,i}.$$

An alternative scale-adjusted measure is

$$\text{Pct Change in Labor Market Concentration} = \frac{\text{Labor Market Concentration}}{\text{Labor Market HHI}}.$$

- **Interpretation.** Only overlapping labor markets contribute to Labor Market Concentration. Nonoverlapping county-$\times$industry cells have one employment share equal to zero and therefore contribute nothing.
- **Why the weighting scheme is sensible.** Overlap in larger labor markets of the merging firms matters more because it affects a larger share of their combined workforce.

2.4 Key descriptive objects

- **Number of deals.** 2,664 completed mergers.
- **Deals with overlap.** 1,126 deals have at least one overlapping labor market.
- **Deal characteristics.**
  - Mean acquirer market capitalization: \$18,997 million.
  - Mean target market capitalization: \$1,389 million.
  - Mean relative size: 0.26.
  - 32% of deals are all-stock.
  - About 1% are hostile.
- **Labor-market concentration statistics (all deals).**
  - Mean Labor Market Concentration: 4.82.
  - Mean Pct Change in Labor Market Concentration: 0.18%.
  - Mean labor market HHI: 2,710.
  - Mean product market HHI: 2,366.
  - Mean percent overlapping employees: 7.08%.
  - Mean percent overlapping sites: 2.28%.
- **Among overlapping deals only.**
  - Mean Labor Market Concentration: 11.41.
  - Median Labor Market Concentration: 0.23.
  - Mean Pct Change in Labor Market Concentration: 0.43%.
  - Mean labor market HHI: 2,781.
  - Mean percent overlapping employees: 16.75%.
  - Mean percent overlapping sites: 5.40%.

2.5 Event windows, related-party portfolios, and postmerger panels

- **Abnormal returns.** Expected returns are estimated with a market model using 240 trading days beginning 300 days before the first successful bid announcement.
- **Event windows.** The paper studies $(-1, +1)$, $(-5, +5)$, and $(-10, +10)$ windows.

- **Combined wealth effect.** The merger wealth effect is the value-weighted abnormal return to acquirer and target:
$$\text{CWE} = w_A \cdot CAR_A + w_T \cdot CAR_T,$$
with weights based on market capitalization 15 trading days before the first successful bid announcement.
- **Rivals.** Rivals are firms with the same primary 4-digit SIC code as the acquirer or target.
- **Customers and suppliers.** Main customer and supplier industries are identified from BEA input-output tables; the paper also studies "key" customers and suppliers disclosed in firm filings.
- **Establishment-level panel.** For the stacked DiD, the treatment group is overlapping county-$\times$industry cells within the merging firms, and the control group is nonoverlapping cells of the same firms. The paper tracks each labor market for three years before and after completion.
- **Firm-level panel.** The authors compare actual merger pairs to pseudo-merger pairs built by matching firms on 2-digit SIC industry and labor market HHI.

3 Models and empirical specifications (all equations + interpretation)

3.1 Competing channels: labor power versus labor efficiency

- **Labor power channel.**
  - More concentrated labor markets give the merged firm more monopsony power.
  - The firm can restrict labor demand and lower wages.
  - Profits rise, but output may fall.
  - Customers and suppliers should be harmed because lower output reduces input demand and raises output prices.
  - Rival firms may benefit if monopsony power spills over through tacit coordination or no-poach-like environments.
- **Labor efficiency channel.**
  - The merged firm reassigns workers, eliminates redundancies, and adopts labor-saving or productivity-enhancing technologies.
  - Output and productivity rise, even if local employment falls in overlapping labor markets.
  - Rivals can be hurt because the merged firm becomes more efficient and more competitive.
  - Customers and suppliers can benefit if output rises or operating efficiency improves.

Interpretation. The central problem in the paper is not whether labor market concentration matters, but *why* it matters.

3.2 Main concentration measures

The paper's two key empirical variables are

$$\text{Labor Market Concentration} = \sum_i \omega_i LShare_{A,i} LShare_{T,i},$$
$$\omega_i = \frac{Emp_{A,i} + Emp_{T,i}}{Emp_A + Emp_T},$$

and

$$\text{Pct Change in Labor Market Concentration} = \frac{\text{Labor Market Concentration}}{\text{Labor Market HHI}}.$$

Interpretation. Labor Market Concentration is the level effect: how much overlap the two firms have in local hiring markets. Pct Change in Labor Market Concentration rescales that overlap by how concentrated those labor markets already were before the deal.

3.3 Baseline merger-value regressions

The baseline multivariate specification takes the form

$$\begin{aligned}\text{CWE}_d = {} & \beta_1 \mathbf{1}\{\text{Labor Market Concentration}_d > 0\} + \beta_2 \text{Labor Market Concentration}_d \\ & + \gamma X_d \\ & + \eta_{\text{ind}} + \eta_t + \varepsilon_d,\end{aligned}$$

An alternative version uses Pct Change in Labor Market Concentration$_d$ instead.

The control vector X_d includes deal and firm characteristics such as:

- headquarters-to-headquarters distance,
- labor market HHI,
- product market HHI,
- acquirer free cash flow,
- an interaction of acquirer free cash flow with a Tobin's-Q indicator,
- hostile-deal status,
- all-stock financing,
- relative deal size,
- target free cash flow,
- target Tobin's Q.

Interpretation. The dummy $\mathbf{1}\{\text{Labor Market Concentration}_d > 0\}$ compares deals with any labor-market overlap to deals with none. The continuous variable Labor Market Concentration$_d$ or Pct Change in Labor Market Concentration$_d$ then asks whether *more* overlap among overlapping deals predicts *more* value creation.

3.4 Heterogeneity by mobility, noncompetes, and right-to-work laws

The paper splits the sample by three labor-market characteristics.

- **Mobility.** High mobility means workers' skills are portable across industries. Labor power predicts stronger effects when mobility is *low*; labor efficiency predicts stronger effects when mobility is *high*.
- **NCA enforceability.** Stronger enforceability of noncompete agreements should make labor power easier to exercise. So labor power predicts stronger merger gains in high-NCA states.
- **Right-to-work laws.** A higher RTW measure proxies for weaker union power. If unions mitigate labor market power or resist efficiency-enhancing reorganization, the split helps identify which force dominates.

Interpretation. These splits are not the main identification device, but they are very useful for channel diagnosis.

3.5 Rival, customer, and supplier wealth-effect regressions

For a related-party portfolio p, the paper estimates regressions of the form

$$
\begin{aligned}
CAR_d^p = {} & \beta_1 \mathbf{1}\{\text{Labor Market Concentration}_d > 0\} + \beta_2 \text{Labor Market Concentration}_d \\
& + \delta \text{CWE}_d + \gamma X_d \\
& + \eta_{\text{ind}} + \eta_t + \varepsilon_d,
\end{aligned}
$$

Again, one can use the percentage-change measure instead.

- Including CWE_d helps isolate whether related-party reactions reflect the market's interpretation of the labor-market mechanism, conditional on overall deal quality.
- The paper also studies a more targeted rival portfolio: rivals operating specifically in the overlapping local labor markets of the deal.

Interpretation. Signs on related-party returns are the cleanest way to distinguish whether the merged firm's gains look more like output reduction and rent extraction, or more like productive efficiency.

3.6 Establishment-level stacked difference-in-differences

The establishment-level specification is

$$
Y_{it} = \beta_0 + \beta_1(\text{Treatment}_i \times Post_t) + FE_{deal \times labor\ market} + FE_{deal \times year} + \varepsilon_{it},
$$

where the treatment variable is alternatively

- $\mathbf{1}$\{Positive local Labor Market Concentration\},
- local Labor Market Concentration, or
- local Pct Change in Labor Market Concentration.

The outcomes are

- $\log(1 + \text{Employees})$, and
- IT budget per employee.

Interpretation. This design compares overlapping and nonoverlapping labor markets *within the same merging firms* before and after the merger. It is therefore much more local than the announcement-return evidence.

3.7 Firm-level stacked difference-in-differences

The firm-level real-effects tests compare actual merger pairs to pseudo-merger pairs:

$$Outcome_{pt} = \beta(Post_t \times MergerPair_p) + FE_{deal \times year} + FE_{deal \times pair} + \varepsilon_{pt}.$$

The paper estimates this specification separately for high- and low-overlap deals, using either Labor Market Concentration or Pct Change in Labor Market Concentration to define the split. The outcomes are

- log(Revenues),
- log(Operating Income),
- Operating Income/Revenues,
- labor productivity (revenues per employee).

Interpretation. If the labor-efficiency story is right, high-overlap mergers should look more productive after completion, not merely more profitable through higher margins.

4 Identification and empirical credibility (what makes the evidence persuasive)

4.1 What the paper can and cannot identify

- The paper identifies how *deal-induced changes in labor market concentration* relate to firm value, related-party reactions, and postmerger real outcomes.
- The paper does *not* claim that labor power is impossible in all mergers. Rather, it asks which channel best explains the *average pattern* in this sample of completed U.S. public mergers.

4.2 Why the labor-market overlap measure is useful

- The key measurement object is built from establishment-level employment and local labor markets rather than only from firm headquarters or product-market overlap.
- This matters because mergers can be close in labor markets even when product-market overlap is limited, and vice versa.
- The measure also includes private-firm employment in the denominator of local shares, making the concentration concept more economically meaningful than one based only on public firms.

4.3 How related-party returns separate channels

- **If labor power dominates:** customers and suppliers should be hurt, because merged firms would restrict output and demand fewer inputs.
- **If labor efficiency dominates:** customers and suppliers can benefit, while rivals may be harmed because the merged firm becomes more productive and competitively tougher.
- The paper's combination of positive customer/supplier reactions and negative local-rival reactions is therefore especially informative.

4.4 How the labor-market splits sharpen interpretation

- Stronger results in **high-mobility** labor markets are hard to reconcile with monopsony power, which should be easier when workers cannot move.
- Stronger results when **NCA enforceability is low** also cut against the labor-power story.
- Weaker effects when union power is stronger are consistent with efficiency being harder to realize when reorganization is constrained.

4.5 What the real-effects evidence adds

- Local employment declines by more in overlapping labor markets, but IT spending rises. That pattern looks like labor-saving reorganization rather than pure output contraction.
- At the firm level, revenues and operating income rise only for high-overlap mergers, operating margins do not rise, and labor productivity increases. That is much more consistent with efficiency than with monopsony-driven profit gains.

4.6 Limits and relation to the literature

- The analysis is about **completed mergers of U.S. public firms**, so it does not speak directly to blocked mergers, private-private deals, or sectors where local labor markets are even narrower.
- The paper's evidence is most closely connected to the M&A literature on value creation and the labor-market-concentration literature on monopsony and antitrust.
- A balanced reading is important: the evidence favors efficiency gains on average, but the authors explicitly note that some workers in overlapping labor markets are likely harmed through displacement.

5 Tables and figures

5.1 Table I: sample and labor-market overlap

Read:

- The paper studies 2,664 completed public-public mergers from 1991 to 2016, of which 1,126 feature at least one overlapping labor market.

Table I

Sample Distribution across Years and Summary Statistics

Panel A shows the distribution of mergers and acquisitions announced between 1991 and 2016. Only acquisitions in which both acquirer and target are public firms and matched to the NETS database are considered. We require that stock return data be available on the CRSP database for both merging firms. Panel B presents summary statistics for our sample of acquisitions. Panel C summarizes labor market characteristics. *ΔLabor Market Concentration* is the deal-level increase in labor market concentration. *Labor Market HHI* (*Product Market HHI*) is the deal-level HHI based on employment (revenues). All variable definitions are provided in the Appendix.

Panel A: Frequency Distribution of Deals in Sample

Year	Num.	Pct.	Year	Num.	Pct.	Year	Num.	Pct.
1991	51	1.91	2000	166	6.23	2009	59	2.21
1992	35	1.31	2001	137	5.14	2010	70	2.63
1993	42	1.58	2002	76	2.85	2011	80	3.00
1994	105	3.94	2003	87	3.27	2012	80	3.00
1995	152	5.71	2004	78	2.93	2013	71	2.67
1996	148	5.56	2005	89	3.34	2014	86	3.23
1997	199	7.47	2006	85	3.19	2015	92	3.45
1998	237	8.9	2007	83	3.12	2016	84	3.15
1999	214	8.03	2008	58	2.18			
						Total	2,664	100.00

Panel B: Summary Statistics for Deal-Level and Firm-Level Attributes

Variable	*N*	Mean	Std. Dev.	p25	p50	p75
Acquirer Characteristics:						
Acquirer Market Capitalization ($ m.)	2,664	18,997.10	51,158.21	550.03	2,252.74	10,661.26
Acquirer FCF	2,664	0.06	0.13	0.03	0.08	0.12
Acquirer Tobin's Q	2,643	2.36	2.82	1.24	1.71	2.61

(*Continued*)

Table I—*Continued*

Panel B: Summary Statistics for Deal-Level and Firm-Level Attributes

Variable	*N*	Mean	Std. Dev.	p25	p50	p75
Target Characteristics:						
Target Market Capitalization ($ m.)	2,664	1,389.13	5,055.43	60.09	205.03	805.67
Target Tobin's Q	2,642	2.02	1.82	1.09	1.47	2.26
Target FCF	2,664	0.001	0.24	−0.002	0.06	0.11
Deal Characteristics:						
Relative Size	2,664	0.26	0.38	0.03	0.12	0.36
Hostile Deal	2,664	0.01	0.12	0.00	0.00	0.00
All Stock Deal	2,664	0.32	0.47	0.00	0.00	1.00
CWE (−1,+1) (%)	2,664	1.91	8.11	−1.91	1.20	5.29
CWE (−5,+5) (%)	2,652	2.03	10.64	−3.22	1.70	7.25
CWE (−10,+10) (%)	2,649	2.01	13.32	−4.61	1.71	8.56

Panel C: Summary Statistics for Labor Market Concentration and Increase in Labor Market Concentration

Variable	*N*	Mean	Std. Dev.	p5	p10	p25	p50	p75	p90	p95
Increase in Labor Market Concentration (for All Deals in Sample):										
ΔLabor Market Concentration	2,664	4.82	25.93	0.00	0.00	0.00	0.00	0.07	5.05	21.23
Pct Change in ΔLabor Market Concentration (%)	2,664	0.18	0.84	0.00	0.00	0.00	0.00	0.004	0.24	0.88
Labor Market HHI	2,664	2,710.32	2,025.73	273.25	445.01	1,013.31	2,282.75	4,065.60	5,800.66	6,675.62
Product Market HHI	2,547	2,366.31	1,832.81	250.84	383.60	866.82	1,890.01	3,498.99	5,057.91	5,948.22
ΔLabor Market Concentration Ratio (Top 4) (%)	2,664	0.14	0.80	0.00	0.00	0.00	0.00	0.002	0.21	0.62
Pct Overlapping Employees (%)	2,664	7.08	16.00	0.00	0.00	0.00	0.00	4.61	26.20	45.43
Pct Overlapping Sites (%)	2,664	2.28	4.74	0.00	0.00	0.00	0.00	2.30	7.29	12.93
Increase in Labor Market Concentration (for Deals with One or More Overlapping Labor Market):										
ΔLabor Market Concentration	1,126	11.41	38.95	0.00001	0.0001	0.01	0.23	4.35	27.63	56.00
Pct Change in ΔLabor Market Concentration (%)	1,126	0.43	1.25	0.000001	0.00001	0.0003	0.013	0.22	1.06	2.18
Labor Market HHI	1,126	2,780.79	1,928.67	304.43	493.26	1,168.27	2,456.33	4,101.92	5,673.70	6,406.83
Product Market HHI	1,118	2,363.20	1,748.97	270.93	439.52	917.92	1,898.53	3,544.45	4,978.35	5,704.34
ΔLabor Market Concentration Ratio (Top 4) (%)	1,126	0.33	1.20	0.00	0.00	0.00	0.014	0.19	0.74	1.40
Pct Overlapping Employees (%)	1,126	16.75	21.08	0.17	0.44	1.90	7.28	23.84	49.99	66.00
Pct Overlapping Sites (%)	1,126	5.40	6.03	0.28	0.50	1.22	3.03	7.04	14.29	20.00

Table II

Average Announcement Period Wealth Effects for the Combined Firm (*CWE*)

The sample comprises all successful acquisitions announced between 1991 and 2016. Average announcement-period wealth effects (*CWE*) for various subsamples are reported. Panel A is based on subsamples of positive versus zero value for the deal-level increase in labor market concentration (*ΔLabor Market Concentration*). Panel B is based on deals with at least one overlapping labor market between merging firms, and subsamples are formed for above- versus below-median *ΔLabor Market Concentration*. *CWE* is the cumulative abnormal return of the acquirer and target firms, weighted by their market capitalization prior to deal announcement. Mean values in the table are percentages, with *t*-statistics in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

Panel A: High vs. Low *ΔLabor Market Concentration* Based on All Deals

	ΔLabor Market Concentration > 0		*ΔLabor Market Concentration* = 0		
Window	*N*	Mean % (*t*-Statistic)	*N*	Mean % (*t*-Statistic)	Difference (*t*-Statistic)
(−1, +1)	1,126	2.26***	1,538	1.65***	0.62*
		(10.00)		(7.63)	(1.93)
(−5, +5)	1,125	2.74***	1,531	1.50***	1.24***
		(9.96)		(5.10)	(2.97)
(−10, +10)	1,124	2.86***	1,529	1.37***	1.49***
		(8.62)		(3.65)	(2.86)

Panel B: High vs. Low *ΔLabor Market Concentration* in Deals with At Least One Overlapping Labor Market

	Above Median *ΔLabor Market Concentration*		Below Median *ΔLabor Market Concentration*		
Window	*N*	Mean % (*t*-Statistic)	*N*	Mean % (*t*-Statistic)	Difference (*t*-Statistic)
(−1, +1)	562	3.15***	564	1.38***	1.77***
		(9.59)		(4.49)	(3.93)
(−5, +5)	563	3.63***	562	1.85***	1.78***
		(9.00)		(4.99)	(3.25)
(−10, +10)	563	3.54***	561	2.19***	1.35**
		(7.89)		(4.47)	(2.03)

- The average labor market HHI is already high: about 2,710 in the full sample and 2,781 among overlapping deals. So the typical labor market in the sample is not especially competitive to begin with.
- The mean Labor Market Concentration is 4.82 in the full sample and 11.41 among overlapping deals, while the median among overlapping deals is only 0.23. This signals substantial right-skewness: a relatively small number of deals generate very large overlap shocks.
- Among overlapping deals, average overlapping employees equal 16.75% of the combined workforce. That is economically meaningful overlap, not just a trivial by-product of diversified firms.

5.2 Table II and Table IV: combined wealth effects

Read:

- Table II gives the paper's headline reduced-form result. In the $(-5, +5)$ window, deals with positive labor-market overlap have average CWE of 2.74%, versus 1.50% for deals with no overlap. Among overlapping deals, above-median overlap deals have 3.63% versus 1.85% for below-median overlap deals.

- Table IV shows these patterns survive a multivariate setting with controls for geography, labor-market structure, product-market concentration, financing method, free cash flow, and deal size.
- The continuous overlap coefficients are positive and significant: for example, the coefficient on Labor Market Concentration in the $(-5, +5)$ window is 0.0139 and significant at 5%; the coefficient on Pct Change in Labor Market Concentration is 0.5927 and significant at 1%.
- Economically, these tables say that merger value is higher when the acquirer and target share more labor markets. But standing alone, this result does *not* tell us whether the market is rewarding labor power or labor efficiency.

Table IV

Determinants of Combined Wealth Effects (*CWE*)

This table presents OLS regressions of combined wealth effects for completed acquisitions. *CWE* is the value-weighted cumulative abnormal return of the acquirer and target firms. *ΔLabor Market Concentration* is the deal-level increase in labor market concentration. *Pct Change in ΔLabor Market Concentration* is the deal-level percentage change in labor market concentration. *Positive ΔLabor Market Concentration* is a dummy variable set to one when *ΔLabor Market Concentration* is positive, and zero otherwise. Other variables are defined in the Appendix. t-Statistics reported in parentheses are based on heteroskedasticity-robust standard errors. ***, ** and * indicate significance at 1%, 5%, and 10% level, respectively.

Dependent Variable	*CWE (−1, +1)*			*CWE (−5, +5)*			*CWE (−10, +10)*		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Positive ΔLabor Market Concentration	0.2043	0.1188	0.0946	0.6281*	0.4799	0.3943	0.8740*	0.7450	0.7063
	(0.68)	(0.39)	(0.31)	(1.65)	(1.25)	(1.02)	(1.81)	(1.52)	(1.43)
ΔLabor Market Concentration		0.0080*			0.0139**			0.0121**	
		(1.73)			(2.35)			(2.15)	
Pct Change in ΔLabor Market Concentration			0.2791*			0.5927***			0.4251*
			(1.72)			(2.66)			(1.86)
HQ to HQ Distance	−0.0003	−0.0003	−0.0003	−0.0002	−0.0002	−0.0001	−0.0005	−0.0005	−0.0005
	(−1.62)	(−1.56)	(−1.51)	(−0.72)	(−0.63)	(−0.54)	(−1.52)	(−1.46)	(−1.42)
Labor Market HHI	−0.0003*	−0.0003*	−0.0003*	−0.0003	−0.0003	−0.0003	−0.0002	−0.0002	−0.0002
	(−1.68)	(−1.70)	(−1.67)	(−1.14)	(−1.16)	(−1.12)	(−0.78)	(−0.80)	(−0.77)
Product Market HHI	0.0004**	0.0004**	0.0004**	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004
	(2.33)	(2.33)	(2.36)	(1.51)	(1.50)	(1.55)	(1.31)	(1.31)	(1.34)
Acquirer FCF	−1.5679	−1.5678	−1.6301	−10.8054	−10.8081	−10.9434	−6.4710	−6.4734	−6.5700
	(−0.22)	(−0.22)	(−0.23)	(−1.13)	(−1.12)	(−1.14)	(−0.39)	(−0.39)	(−0.40)
Acquirer FCF × Tobin's Q dummy	3.4779	3.4347	3.4701	12.7338	12.6629	12.7251	7.9279	7.8662	7.9217
	(0.45)	(0.44)	(0.45)	(1.22)	(1.21)	(1.21)	(0.46)	(0.45)	(0.46)
Hostile Deal	1.9553	1.9301	1.9185	0.5162	0.4658	0.4312	−0.2477	−0.2916	−0.3087
	(1.42)	(1.40)	(1.39)	(0.35)	(0.31)	(0.29)	(−0.14)	(−0.16)	(−0.17)
All Stock Deal	−2.7005***	−2.7058***	−2.7022***	−2.9102***	−2.9195***	−2.9133***	−2.5922***	−2.6003***	−2.5944***
	(−6.89)	(−6.90)	(−6.90)	(−5.46)	(−5.48)	(−5.48)	(−3.77)	(−3.78)	(−3.78)

(*Continued*)

Table IV—*Continued*

Dependent Variable	*CWE (−1, +1)*			*CWE (−5, +5)*			*CWE (−10, +10)*		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Relative Size	4.5223***	4.4646***	4.4627***	5.4835***	5.3833***	5.3565***	5.5273***	5.4400***	5.4362***
	(5.89)	(5.80)	(5.79)	(5.70)	(5.62)	(5.58)	(5.20)	(5.11)	(5.10)
Target FCF	−0.8826	−0.8693	−0.8712	−1.5609	−1.5382	−1.5370	−1.2555	−1.2358	−1.2384
	(−1.06)	(−1.04)	(−1.04)	(−1.25)	(−1.23)	(−1.23)	(−0.92)	(−0.90)	(−0.91)
Target Tobin's Q	−0.3282**	−0.3266**	−0.3254**	−0.4336**	−0.4309**	−0.4276**	−0.6546***	−0.6522***	−0.6503***
	(−2.28)	(−2.27)	(−2.27)	(−2.37)	(−2.37)	(−2.35)	(−2.86)	(−2.86)	(−2.85)
Industry and year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,530	2,530	2,530	2,523	2,523	2,523	2,523	2,523	2,523
R^2	0.124	0.124	0.124	0.104	0.105	0.106	0.087	0.087	0.087

5.3 Table V: mobility, noncompetes, and unions

Table V

Effect of Worker Mobility, Noncompete Agreement (NCA) Enforceability, and Right-to-Work (RTW) Laws

This table presents OLS regressions of combined wealth effects for completed acquisitions announced between 1991 and 2016. The dependent variable is combined wealth effect of merging firms (*CWE*). The high and low subsamples are formed based on the sample median value for the variables *Mobility*, *NCA*, and *RTW*. All variables are defined in the Appendix. t-Statistics reported in the parentheses are based on heteroskedasticity-robust standard errors. ***, **, and * indicate significance at 1%, 5%, and 10% level, respectively.

Dep. Variable:	*CWE (−5, +5)*				*CWE (−5, +5)*				*CWE (−5, +5)*			
	(1) High *Mobility*	(2) Low *Mobility*	(3) High *Mobility*	(4) Low *Mobility*	(5) High *NCA*	(6) Low *NCA*	(7) High *NCA*	(8) Low *NCA*	(9) High *RTW*	(10) Low *RTW*	(11) High *RTW*	(12) Low *RTW*
ΔLabor Market Concentration	0.0156**	0.0117			0.0123	0.0223***			0.0265***	0.0031		
	(2.07)	(1.18)			(1.52)	(2.74)			(2.65)	(0.43)		
Pct Change in ΔLabor Market Concentration			0.9231**	0.4917**			0.7624**	0.8363***			1.1209***	0.3641
			(2.49)	(1.98)			(2.02)	(4.11)			(3.48)	(1.57)
Positive ΔLabor Market Concentration	0.5373	0.5292	0.3753	0.4505	0.3739	0.6654	0.2304	0.5469	0.2691	0.6753	0.1404	0.5684
	(1.04)	(0.91)	(0.72)	(0.77)	(0.72)	(1.02)	(0.45)	(0.83)	(0.47)	(1.21)	(0.24)	(1.01)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry and year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,271	1,237	1,271	1,237	1,200	1,159	1,200	1,159	1,235	1,204	1,235	1,204
R^2	0.115	0.129	0.118	0.130	0.135	0.110	0.138	0.112	0.158	0.085	0.162	0.086

Table III

Wealth Effects for Product Market Rivals, Suppliers, and Customers by *ΔLabor Market Concentration*

This table presents average announcement period wealth effects for subsamples of high versus low *ΔLabor Market Concentration*. *Acquirer Rival CAR EW* and *Target Rival CAR EW* are the cumulative abnormal return of an equal-weighted portfolio of the acquirer and target rival firms, respectively. *All Rival CAR EW* is the cumulative abnormal return of an equal-weighted portfolio of all rivals of the merging firms. *Acquirer Customer CAR EW* and *Target Customer CAR EW* are the cumulative abnormal return of an equal-weighted portfolio of the acquirer and target main customer industry firms, respectively. *All Customer CAR EW* is the cumulative abnormal return of an equal-weighted portfolio of firms in the main customer industries of both the acquirer and target. *Acquirer Supplier CAR EW* and *Target Supplier CAR EW* are the cumulative abnormal return of an equal-weighted portfolio of the acquirer and target main supplier industry firms, respectively. *All Supplier CAR EW* is the cumulative abnormal return of an equal-weighted portfolio of firms in the main supplier industry of both the acquirer and target. Mean values in table are percentages, with t-statistics in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

Panel A: Announcement Period Abnormal Returns to Portfolios of Rival Firms

	ΔLabor Market Concentration > 0		*ΔLabor Market Concentration* = 0		
	N	Mean % (t-Statistic)	*N*	Mean % (t-Statistic)	Difference (t-Statistic)
Acquirer Rival CAR EW (−5, +5)	1,010	0.27*	1,488	0.11	0.16
		(1.86)		(0.84)	(0.85)
Acquirer Rival CAR EW (−10, +10)	1,010	0.37*	1,488	0.41**	−0.04
		(1.76)		(1.96)	(−0.10)
Target Rival CAR EW (−5, +5)	1,026	0.64***	1,488	0.34***	0.31
		(4.67)		(2.53)	(1.55)
Target Rival CAR EW (−10, +10)	1,026	0.75***	1,488	0.35	0.40
		(3.72)		(1.61)	(1.30)
All Rival CAR EW (−5, +5)	1,052	0.49***	1,524	0.21*	0.28
		(3.84)		(1.86)	(1.55)
All Rival CAR EW (−10, +10)	1,052	0.58***	1,524	0.36*	0.22
		(3.03)		(1.87)	(0.80)

Panel B: Announcement Period Abnormal Returns to Portfolios of Customer Firms

	ΔLabor Market Concentration > 0		*ΔLabor Market Concentration* = 0		
	N	Mean % (t-Statistic)	*N*	Mean % (t-Statistic)	Difference (t-Statistic)
Acquirer Customer CAR EW (−5, +5)	816	0.20	1,222	−0.13	0.33
		(0.99)		(−1.10)	(1.50)
Acquirer Customer CAR EW (−10, +10)	816	0.41	1,222	−0.12	0.54*
		(1.54)		(−0.70)	(1.75)
Target Customer CAR EW (−5, +5)	794	0.40**	1,194	−0.13	0.53**
		(1.98)		(−1.00)	(2.35)

(*Continued*)

Table III—*Continued*

Panel B: Announcement Period Abnormal Returns to Portfolios of Customer Firms

	ΔLabor Market Concentration > 0		*ΔLabor Market Concentration* = 0		
	N	Mean % (t-Statistic)	*N*	Mean % (t-Statistic)	Difference (t-Statistic)
Target Customer CAR EW (−10, +10)	794	0.77***	1,194	0.09	0.68**
		(2.74)		(0.45)	(2.10)
All Customer CAR EW (−5, +5)	888	0.28	1,329	−0.10	0.38*
		(1.52)		(−0.92)	(1.90)
All Customer CAR EW (−10, +10)	888	0.62**	1,329	−0.003	0.62**
		(2.43)		(−0.015)	(2.15)

Panel C: Announcement Period Abnormal Returns to Portfolios of Supplier Firms

	ΔLabor Market Concentration > 0		*ΔLabor Market Concentration* = 0		
	N	Mean % (t-Statistic)	*N*	Mean % (t-Statistic)	Difference (t-Statistic)
Acquirer Supplier CAR EW (−5, +5)	813	0.44***	1,135	−0.21*	0.65***
		(2.94)		(−1.70)	(3.35)
Acquirer Supplier CAR EW (−10, +10)	813	0.53**	1,135	−0.27	0.80***
		(2.37)		(−1.44)	(2.75)
Target Supplier CAR EW (−5, +5)	809	0.37**	1,124	−0.05	0.42**
		(2.65)		(−0.40)	(2.20)
Target Supplier CAR EW (−10, +10)	809	0.57***	1,124	−0.03	0.60**
		(2.79)		(−0.18)	(2.20)
All Supplier CAR EW (−5, +5)	870	0.51***	1,234	−0.10	0.61***
		(3.70)		(−0.91)	(3.50)
All Supplier CAR EW (−10, +10)	870	0.73***	1,234	−0.10	0.83***
		(3.61)		(−.62)	(3.25)

Read:

- The relation between labor-market overlap and merger gains is stronger in **high-mobility** environments. This is exactly the opposite of what a simple monopsony story would predict.

- The relation is also stronger when **noncompete agreements are less enforceable**. Again, that cuts against labor power, which should be easier when worker mobility is restricted.
- The effect is concentrated in states with **weaker union power** (high right-to-work exposure). The paper interprets this as unions muting the impact of increased labor-market concentration, either because they countervail labor power or because they resist efficiency-enhancing restructuring.
- Table V is one of the paper's most important channel tables, because it uses economically motivated state variables to discriminate between stories rather than merely documenting average treatment effects.

5.4 Table III, Table VI, and Table VII: rival-firm reactions

Table VI

Determinants of Wealth Effects of Rival Firms

This table presents OLS regressions of wealth effects of rival firms in completed acquisitions announced between 1991 and 2016. The dependent variable in Panels A and B is *Acquirer Rival CAR EW* and *Target Rival CAR EW*, respectively. In Panel C, the dependent variable is *All Rival CAR EW*. Control variables include those identified in equation (3) as well as the combined wealth effect to merging firms (*CWE*). All variables are defined in the Appendix. *t*-Statistics reported in the parentheses are based on heteroskedasticity- robust standard errors. ***, **, and * indicate significance at 1%, 5%, and 10% level, respectively.

Panel A: Determinants of Acquirer Rival Wealth Effects (*Acquirer Rival CAR EW*)

Dependent Variable	*Acquirer Rival CAR EW*					
	(−5, +5) (1)	(−5, +5) (2)	(−5, +5) (3)	(−10, +10) (4)	(−10, +10) (5)	(−10, +10) (6)
Positive ΔLabor Market Concentration	0.2384 (1.23)	0.3260* (1.66)	0.3173 (1.60)	−0.0423 (−0.14)	0.0464 (0.16)	0.0469 (0.16)
ΔLabor Market Concentration		−0.0088** (−2.47)			−0.0089* (−1.71)	
Pct Change in ΔLabor Market Concentration			−0.2105** (−2.04)			−0.2377* (−1.74)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry and year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,310	2,310	2,310	2,310	2,310	2,310
R^2	0.090	0.092	0.091	0.114	0.115	0.114

Panel B: Determinants of Target Rival Wealth Effects (*Target Rival CAR EW*)

Dependent Variable	*Target rival CAR EW*					
	(−5, +5) (1)	(−5, +5) (2)	(−5, +5) (3)	(−10, +10) (4)	(−10, +10) (5)	(−10, +10) (6)
Positive ΔLabor Market Concentration	0.2758 (1.38)	0.3322 (1.64)	0.3195 (1.57)	0.3472 (1.16)	0.4167 (1.37)	0.4132 (1.36)
ΔLabor Market Concentration		−0.0057* (−1.78)			−0.0070 (−1.37)	
Pct Change in ΔLabor Market Concentration			−0.1177 (−1.02)			−0.1769 (−1.36)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry and year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,325	2,325	2,325	2,325	2,325	2,325
R^2	0.075	0.076	0.075	0.109	0.109	0.109

(*Continued*)

Table VI—*Continued*

Panel C: Determinants of Acquirer and Target Rival Wealth Effects (*All Rival CAR EW*)

Dependent Variable	*All Rival CAR EW*					
	(−5, +5) (1)	(−5, +5) (2)	(−5, +5) (3)	(−10, +10) (4)	(−10, +10) (5)	(−10, +10) (6)
Positive ΔLabor Market Concentration	0.2870* (1.65)	0.3734** (2.11)	0.3653** (2.05)	0.2062 (0.77)	0.2920 (1.08)	0.2980 (1.10)
ΔLabor Market Concentration		−0.0085** (−2.55)			−0.0084* (−1.76)	
Pct Change in ΔLabor Market Concentration			−0.2082** (−2.02)			−0.2433* (−1.87)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry and year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,385	2,385	2,385	2,385	2,385	2,385
R^2	0.091	0.094	0.093	0.127	0.128	0.128

Read:

- Table III's univariate evidence for rivals is mixed. Simple rival portfolios are not enough to cleanly separate the two channels.
- Table VI adds controls and still finds negative rival reactions. For all-rival equal-weighted portfolios, the coefficient on Labor Market Concentration is −0.0085 in the (−5, +5) window and −0.0084 in the (−10, +10) window.
- Table VII is sharper because it looks only at rivals that operate in the *same overlapping labor markets*. There the effects are clearly negative: for all local-labor-market rivals, the coefficient on Labor Market Concentration is −0.0143 in the (−5, +5) window and −0.0280 in the (−10, +10) window.

- This is strong evidence for the labor-efficiency channel. If the merged firm becomes more efficient in precisely the labor markets where overlap increases, nearby rivals should be hurt - and that is what Table VII shows.

Table VII—*Continued*

Panel C: Determinants of *All Labor Market Rival CAR EW*				
Dependent Variable	*All Labor Market Rival CAR EW*			
	(−5, +5) (1)	(−5, +5) (2)	(−10, +10) (3)	(−10, +10) (4)
ΔLabor Market Concentration	−0.0143** (−2.57)		−0.0280*** (−3.27)	
Pct Change in ΔLabor Market Concentration		−0.2976** (−2.23)		−0.5339** (−2.58)
Control variables	Yes	Yes	Yes	Yes
Industry and year dummies	Yes	Yes	Yes	Yes
Observations	880	880	880	880
R^2	0.111	0.110	0.149	0.146

Table VII—*Continued*

Panel C: Determinants of *All Labor Market Rival CAR EW*				
Dependent Variable	*All Labor Market Rival CAR EW*			
	(−5, +5) (1)	(−5, +5) (2)	(−10, +10) (3)	(−10, +10) (4)
ΔLabor Market Concentration	−0.0143** (−2.57)		−0.0280*** (−3.27)	
Pct Change in ΔLabor Market Concentration		−0.2976** (−2.23)		−0.5339** (−2.58)
Control variables	Yes	Yes	Yes	Yes
Industry and year dummies	Yes	Yes	Yes	Yes
Observations	880	880	880	880
R^2	0.111	0.110	0.149	0.146

5.5 Table III, Table VIII, and Table IX: customers and suppliers

- Table III's univariate evidence already points in the direction of labor efficiency: customer and supplier portfolios earn more positive abnormal returns when labor-market overlap is positive.
- Table VIII shows that customer firms generally benefit when the deal has positive labor-market overlap. The positive-overlap dummy is especially strong for target-customer portfolios and for broader all-customer portfolios.
- The continuous overlap coefficients in Table VIII are often statistically weak, but the key-customer tests are still informative: the customers most exposed to the merging firms are not harmed; if anything, they benefit.
- Table IX gives the analogous result for suppliers. Supplier returns are positive when labor-market overlap is positive, and key-supplier returns are increasing in Labor Market Concentration in some specifications.
- These tables are difficult to reconcile with monopsony-based output contraction. If the merged firm were simply squeezing workers and reducing output, customers and suppliers should not be gaining.

Read:

5.6 Table X: establishment-level employment and IT

Read:

- Table X is the paper's cleanest local real-effects evidence. Relative to nonoverlapping labor markets within the same merging firms, overlapping labor markets see a 10.4% decline in employment after the merger.
- At the same time, IT spending per employee rises by about $876 in the baseline specification.
- The continuous-treatment specifications tell the same story: greater local overlap predicts both a larger employment decline and a larger increase in IT spending.
- This is exactly the pattern one would expect from restructuring and capital-labor substitution. The merged firm seems to use overlap to remove redundancies and raise efficiency in the affected labor markets.

Table III

Wealth Effects for Product Market Rivals, Suppliers, and Customers by *ΔLabor Market Concentration*

This table presents average announcement period wealth effects for subsamples of high versus low *ΔLabor Market Concentration*. *Acquirer Rival CAR EW* and *Target Rival CAR EW* are the cumulative abnormal return of an equal-weighted portfolio of the acquirer and target rival firms, respectively. *All Rival CAR EW* is the cumulative abnormal return of an equal-weighted portfolio of all rivals of the merging firms. *Acquirer Customer CAR EW* and *Target Customer CAR EW* are the cumulative abnormal return of an equal-weighted portfolio of the acquirer and target main customer industry firms, respectively. *All Customer CAR EW* is the cumulative abnormal return of an equal-weighted portfolio of firms in the main customer industries of both the acquirer and target. *Acquirer Supplier CAR EW* and *Target Supplier CAR EW* are the cumulative abnormal return of an equal-weighted portfolio of the acquirer and target main supplier industry firms, respectively. *All Supplier CAR EW* is the cumulative abnormal return of an equal-weighted portfolio of firms in the main supplier industry of both the acquirer and target. Mean values in table are percentages, with t-statistics in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

Panel A: Announcement Period Abnormal Returns to Portfolios of Rival Firms

	ΔLabor Market Concentration > 0		*ΔLabor Market Concentration* = 0		
	N	Mean % (t-Statistic)	N	Mean % (t-Statistic)	Difference (t-Statistic)
Acquirer Rival CAR EW (−5, +5)	1,010	0.27*	1,488	0.11	0.16
		(1.86)		(0.84)	(0.85)
Acquirer Rival CAR EW (−10, +10)	1,010	0.37*	1,488	0.41**	−0.04
		(1.76)		(1.96)	(−0.10)
Target Rival CAR EW (−5, +5)	1,026	0.64***	1,488	0.34***	0.31
		(4.67)		(2.53)	(1.55)
Target Rival CAR EW (−10, +10)	1,026	0.75***	1,488	0.35	0.40
		(3.72)		(1.61)	(1.30)
All Rival CAR EW (−5, +5)	1,052	0.49***	1,524	0.21*	0.28
		(3.84)		(1.86)	(1.55)
All Rival CAR EW (−10, +10)	1,052	0.58***	1,524	0.36*	0.22
		(3.03)		(1.87)	(0.80)

Panel B: Announcement Period Abnormal Returns to Portfolios of Customer Firms

	ΔLabor Market Concentration > 0		*ΔLabor Market Concentration* = 0		
	N	Mean % (t-Statistic)	N	Mean % (t-Statistic)	Difference (t-Statistic)
Acquirer Customer CAR EW (−5, +5)	816	0.20	1,222	−0.13	0.33
		(0.99)		(−1.10)	(1.50)
Acquirer Customer CAR EW (−10, +10)	816	0.41	1,222	−0.12	0.54*
		(1.54)		(−0.70)	(1.75)
Target Customer CAR EW (−5, +5)	794	0.40**	1,194	−0.13	0.53**
		(1.98)		(−1.00)	(2.35)

(*Continued*)

Table III—*Continued*

Panel B: Announcement Period Abnormal Returns to Portfolios of Customer Firms

	ΔLabor Market Concentration > 0		*ΔLabor Market Concentration* = 0		
	N	Mean % (t-Statistic)	N	Mean % (t-Statistic)	Difference (t-Statistic)
Target Customer CAR EW (−10, +10)	794	0.77***	1,194	0.09	0.68**
		(2.74)		(0.45)	(2.10)
All Customer CAR EW (−5, +5)	888	0.28	1,329	−0.10	0.38*
		(1.52)		(−0.92)	(1.90)
All Customer CAR EW (−10, +10)	888	0.62**	1,329	−0.003	0.62**
		(2.43)		(−0.015)	(2.15)

Panel C: Announcement Period Abnormal Returns to Portfolios of Supplier Firms

	ΔLabor Market Concentration > 0		*ΔLabor Market Concentration* = 0		
	N	Mean % (t-Statistic)	N	Mean % (t-Statistic)	Difference (t-Statistic)
Acquirer Supplier CAR EW (−5, +5)	813	0.44***	1,135	−0.21*	0.65***
		(2.94)		(−1.70)	(3.35)
Acquirer Supplier CAR EW (−10, +10)	813	0.53**	1,135	−0.27	0.80***
		(2.37)		(−1.44)	(2.75)
Target Supplier CAR EW (−5, +5)	809	0.37**	1,124	−0.05	0.42**
		(2.65)		(−0.40)	(2.20)
Target Supplier CAR EW (−10, +10)	809	0.57***	1,124	−0.03	0.60**
		(2.79)		(−0.18)	(2.20)
All Supplier CAR EW (−5, +5)	870	0.51***	1,234	−0.10	0.61***
		(3.70)		(−0.91)	(3.50)
All Supplier CAR EW (−10, +10)	870	0.73***	1,234	−0.10	0.83***
		(3.61)		(−.62)	(3.25)

Table VIII

Determinants of Wealth Effects of Customer Firms

This table presents OLS regressions of combined wealth effects of customer firms in completed acquisitions announced between 1991 and 2016. The dependent variable in Panels A and B is *Acquirer Customer CAR EW* and *Target Customer CAR EW*, respectively. In Panel C, the dependent variable is *All Customer CAR EW* and in Panel D the dependent variable is *Key Customer CAR EW*. Control variables include those identified in equation (3) as well as the combined wealth effect to merging firms (*CWE*). All variables are defined in the Appendix. t-Statistics reported in the parentheses are based on heteroskedasticity-robust standard errors. ***, **, and * indicate significance at 1%, 5%, and 10% level, respectively.

Panel A: Determinants of Acquirer Customer Wealth Effects (*Acquirer Customer CAR EW*)

Dependent Variable	*Acquirer Customer CAR EW*					
	(−5, +5) (1)	(−5, +5) (2)	(−5, +5) (3)	(−10, +10) (4)	(−10, +10) (5)	(−10, +10) (6)
Positive ΔLabor Market Concentration	0.1846	0.2120	0.1912	0.4956	0.5689*	0.5306
	(0.77)	(0.85)	(0.75)	(1.51)	(1.67)	(1.54)
ΔLabor Market Concentration		−0.0030			−0.0081	
		(−0.92)			(−1.24)	
Pct Change in ΔLabor Market Concentration			−0.0203			−0.1074
			(−0.19)			(−0.55)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry and year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,875	1,875	1,875	1,875	1,875	1,875
R^2	0.056	0.056	0.056	0.079	0.080	0.079

Panel B: Determinants of Target Customer Wealth Effects (*Target Customer CAR EW*)

Dependent Variable	*Target Customer CAR EW*					
	(−5, +5) (1)	(−5, +5) (2)	(−5, +5) (3)	(−10, +10) (4)	(−10, +10) (5)	(−10, +10) (6)
Positive ΔLabor Market Concentration	0.4646*	0.4862*	0.4565*	0.7153**	0.7599**	0.7293**
	(1.93)	(1.95)	(1.80)	(2.14)	(2.21)	(2.10)
ΔLabor Market Concentration		−0.0024			−0.0050	
		(−0.77)			(−0.84)	

(*Continued*)

Table VIII—*Continued*

Panel B: Determinants of Target Customer Wealth Effects (*Target Customer CAR EW*)

Dependent Variable	*Target Customer CAR EW*					
	(−5, +5) (1)	(−5, +5) (2)	(−5, +5) (3)	(−10, +10) (4)	(−10, +10) (5)	(−10, +10) (6)
Pct Change in ΔLabor Market Concentration			0.0248			−0.0429
			(0.22)			(−0.26)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry and year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,831	1,831	1,831	1,831	1,831	1,831
R^2	0.079	0.079	0.079	0.104	0.104	0.104

Panel C: Determinants of Acquirer and Target Customer Wealth Effects (*All Customer CAR EW*)

Dependent Variable	*All Customer CAR EW*					
	(−5, +5) (1)	(−5, +5) (2)	(−5, +5) (3)	(−10, +10) (4)	(−10, +10) (5)	(−10, +10) (6)
Positive ΔLabor Market Concentration	0.2584	0.2804	0.2462	0.6003**	0.6663**	0.6255**
	(1.19)	(1.24)	(1.08)	(1.97)	(2.13)	(1.99)
ΔLabor Market Concentration		−0.0025			−0.0075	
		(−0.75)			(−1.16)	
Pct Change in ΔLabor Market Concentration			0.0380			−0.0782
			(0.34)			(−0.44)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry and year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,046	2,046	2,046	2,046	2,046	2,046
R^2	0.065	0.065	0.065	0.092	0.093	0.092

(*Continued*)

Table VIII—*Continued*

Panel D: Determinants of Key Customer Wealth Effects (*Key Customer CAR EW*)

Dependent Variable	*Key customer CAR EW*					
	(−5, +5) (1)	(−5, +5) (2)	(−5, +5) (3)	(−10, +10) (4)	(−10, +10) (5)	(−10, +10) (6)
Positive ΔLabor Market Concentration	0.0130	−0.1089	−0.1351	−0.6109	−0.7280	−0.7727
	(0.03)	(−0.25)	(−0.31)	(−0.98)	(−1.16)	(−1.23)
ΔLabor Market Concentration		0.0155***			0.0148**	
		(2.88)			(1.97)	
Pct Change in ΔLabor Market Concentration			0.5592***			0.6093**
			(2.64)			(2.50)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry and year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,027	1,027	1,027	1,027	1,027	1,027
R^2	0.050	0.053	0.053	0.065	0.066	0.067

Table IX

Determinants of Wealth Effects of Supplier Firms

This table presents OLS regressions of wealth effects of customer firms in completed acquisitions announced between 1991 and 2016. The dependent variable in Panels A and B is *Acquirer Supplier CAR EW* and *Target Supplier CAR EW*, respectively. In Panel C, the dependent variable is *All Supplier CAR EW* and in Panel D the dependent variable is *Key Supplier CAR EW*. Control variables include those identified in equation (3) as well as the combined wealth effect to merging firms (*CWE*). All variables are defined in the Appendix. t-Statistics reported in the parentheses are based on heteroskedasticity-robust standard errors. ***, **, and * indicate significance at 1%, 5%, and 10% level, respectively.

Panel A: Determinants of Acquirer Supplier Wealth Effects (*Acquirer Supplier CAR EW*)

Dependent Variable	*Acquirer Supplier CAR EW*					
	(−5, +5) (1)	(−5, +5) (2)	(−5, +5) (3)	(−10, +10) (4)	(−10, +10) (5)	(−10, +10) (6)
Positive ΔLabor Market Concentration	0.6032***	0.6414***	0.6286***	0.8090***	0.8706***	0.8356***
	(2.92)	(3.06)	(2.96)	(2.69)	(2.85)	(2.70)
ΔLabor Market Concentration		−0.0033			−0.0054	
		(−0.94)			(−0.93)	
Pct Change in ΔLabor Market Concentration			−0.0612			−0.0640
			(−0.62)			(−0.43)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry and year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,785	1,785	1,785	1,785	1,785	1,785
R^2	0.077	0.078	0.078	0.110	0.111	0.110

Panel B: Determinants of Target Supplier Wealth Effects (*Target Supplier CAR EW*)

Dependent Variable	*Target Supplier CAR EW*					
	(−5, +5) (1)	(−5, +5) (2)	(−5, +5) (3)	(−10, +10) (4)	(−10, +10) (5)	(−10, +10) (6)
Positive ΔLabor Market Concentration	0.3618*	0.3573*	0.3386*	0.6035**	0.5352*	0.4887*
	(1.81)	(1.78)	(1.67)	(2.12)	(1.86)	(1.69)
ΔLabor Market Concentration		0.0004			0.0060	
		(0.09)			(0.78)	

(*Continued*)

Table IX—*Continued*

Panel B: Determinants of Target Supplier Wealth Effects (*Target Supplier CAR EW*)

Dependent Variable	*Target Supplier CAR EW*					
	(−5, +5) (1)	(−5, +5) (2)	(−5, +5) (3)	(−10, +10) (4)	(−10, +10) (5)	(−10, +10) (6)
Pct Change in ΔLabor Market Concentration			0.0556			0.2744
			(0.48)			(1.50)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry and year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,770	1,770	1,770	1,770	1,770	1,770
R^2	0.069	0.069	0.069	0.092	0.093	0.094

Panel C: Determinants of Acquirer and Target Supplier Wealth Effects (*All Supplier CAR EW*)

Dependent Variable	*All Supplier CAR EW*					
	(−5, +5) (1)	(−5, +5) (2)	(−5, +5) (3)	(−10, +10) (4)	(−10, +10) (5)	(−10, +10) (6)
Positive ΔLabor Market Concentration	0.5760***	0.5867***	0.5700***	0.8769***	0.8331***	0.7962***
	(3.12)	(3.14)	(3.02)	(3.29)	(3.08)	(2.93)
ΔLabor Market Concentration		−0.0010			0.0039	
		(-0.23)			(0.57)	
Pct Change in ΔLabor Market Concentration			0.0148			0.1981
			(0.13)			(1.18)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry and year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,935	1,935	1,935	1,935	1,935	1,935
R^2	0.087	0.087	0.087	0.118	0.119	0.119

(*Continued*)

Table IX—*Continued*

Panel D: Determinants of Key Supplier Wealth Effects (*Key Supplier CAR EW*)

Panel D: Dependent Variable	*Key Supplier CAR EW*					
	(−5, +5) (1)	(−5, +5) (2)	(−5, +5) (3)	(−10, +10) (4)	(−10, +10) (5)	(−10, +10) (6)
Positive ΔLabor Market Concentration	1.2822*	1.1097	1.2626*	0.8153	0.5045	0.6706
	(1.85)	(1.56)	(1.78)	(0.88)	(0.54)	(0.72)
ΔLabor Market Concentration		0.0232*			0.0412**	
		(1.92)			(2.01)	
Pct Change in ΔLabor Market Concentration			0.0665			0.4892
			(0.17)			(0.89)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry and year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,101	1,101	1,101	1,101	1,101	1,101
R^2	0.059	0.061	0.059	0.068	0.070	0.068

5.7 Table XI: firm-level revenues, operating income, margins, and productivity

Read:

- Table XI moves from local labor markets to merged-firm outcomes. The paper compares actual merger pairs to pseudo-merger pairs and asks whether high-overlap mergers look different after completion.
- For high-overlap mergers, revenues rise and operating income rises. For example, in Panel A, the post-merger-pair coefficient is 0.1896 for log revenues and 0.2839 for log operating income; in Panel B the corresponding coefficients are 0.2061 and 0.2778.
- Operating margins do *not* rise in the high-overlap group, and in the low-overlap group they even fall. That is not the pattern one expects from pure labor power.
- Labor productivity rises only in high-overlap mergers: about 140 in Panel A and 125 in Panel B. Together with the absence of higher margins, this makes the firm-level evidence look much more like productivity improvement than rent extraction.

Table X

Establishment-Level Effects on Employment and IT Spending

This table presents OLS regressions based on completed acquisitions announced between 1991 and 2016. *Log (Employees)* is the logarithm of one plus combined establishment-level employment for the acquirer and target firm. *IT Budget per employee* is the combined establishment-level IT budget for the acquirer and target firm per employee. Results are presented for *Log (Employees)* and *IT Budget per employee* measured in the period $(t-3, t+3)$ years, where year t is the merger completion year. *ΔLabor Market Concentration Est.* is the establishment level increase in labor market concentration. *Positive ΔLabor Market Concentration Est.* is a dummy variable set to one for establishments with a positive value for *ΔLabor Market Concentration Est.*, and zero otherwise. *Post* is a dummy variable set to one for postcompletion years. t-Statistics reported in the parentheses are based on heteroskedasticity-robust standard errors and are clustered by firm-labor market. ***, **, and * indicate significance at 1%, 5%, and 10% level, respectively.

Dependent Variable	*Log (Employees)*			*IT Budget Per Employee*		
	(1)	(2)	(3)	(4)	(5)	(6)
Positive ΔLabor Market Concentration Est. × Post	−0.1040***			875.8689***		
	(−13.55)			(8.18)		
ΔLabor Market Concentration Est. × Post		−1.9488***			10,199.6707***	
		(−9.84)			(2.65)	
Pct Change in ΔLabor Market Concentration × Post			−0.6640***			3,265.6357***
			(−14.45)			(4.21)
Deal × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Deal × Labor Market FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.944	0.944	0.944	0.822	0.822	0.822
Observations	1,647,765	1,647,765	1,647,765	258,002	258,002	258,002

Table XI

Firm-Level Real Effects Tests

This table presents OLS regressions for completed acquisitions announced between 1991 and 2016. Only deals with some overlap in labor markets of merging firms are considered. High and low subsamples are based on above versus below median value for *ΔLabor Market Concentration* (*Pct Change in ΔLabor Market Concentration*) in Panel A (B). *Merger pair* is a dummy set to one for actual pairs of acquirers and targets and zero for pseudo pairs. To form pseudo pairs, we match up to three firms on Compustat for each acquirer and target in a deal based on the two-digit SIC industry and *Labor Market HHI*. We then form pseudo pairs of acquirers and targets based on the matched firms. *Post* is a dummy set to one for years $(t+1)$, $(t+2)$, and $(t+3)$ and zero for years $(t-1)$, $(t-2)$, and $(t-3)$ where year t is the merger completion year. *ΔLabor Market Concentration* is the deal-level increase in labor market concentration. All other variables defined in the Appendix. Estimations contain deal × year and deal × pair fixed effects. t-Statistics reported in parentheses are based on heteroskedasticity-robust standard errors and clustered by firm. ***, **, and * indicate significance at 1%, 5%, and 10% level, respectively.

Panel A: High and Low Groups Formed Based on *ΔLabor Market Concentration*

Dependent Variable:	*Log (Revenues)*		*Log (Operating Income)*		*Operating Income / Revenues*		*Labor Productivity*	
	(1) High	(2) Low	(3) High	(4) Low	(5) High	(6) Low	(7) High	(8) Low
Post × Merger Pair	0.1896*	−0.0076	0.2839**	−0.0590	0.0139	−0.0692*	140.2987*	−19.5572
	(1.71)	(−0.04)	(2.16)	(−0.31)	(0.91)	(−1.78)	(1.75)	(−0.72)
Deal × year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Deal × pair fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	25,534	24,456	23,455	20,460	25,531	24,432	23,314	21,967
R^2	0.991	0.987	0.974	0.962	0.881	0.862	0.957	0.963

Panel B: High and Low Groups Formed Based on *Pct Change in ΔLabor Market Concentration*

Dependent Variable:	*Log (Revenues)*		*Log (Operating Income)*		*Operating Income / Revenues*		*Labor Productivity*	
	(1) High	(2) Low	(3) High	(4) Low	(5) High	(6) Low	(7) High	(8) Low
Post × Merger Pair	0.2061**	−0.0639	0.2778**	−0.1011	0.0138	−0.0811*	124.9301*	−24.0071
	(2.03)	(−0.34)	(2.31)	(−0.48)	(0.99)	(−1.86)	(1.71)	(−0.75)
Deal × year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Deal × pair fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	25,529	24,461	23,192	20,723	25,525	24,438	23,223	22,058
R^2	0.990	0.988	0.973	0.964	0.874	0.866	0.95	0.963

6 Short summary

- **Method.** The paper measures how much each merger increases labor-market concentration using establishment-level employment shares in county-×SIC3 labor markets, and then connects that measure to announcement returns and postmerger real outcomes.
- **Identification logic.** The key empirical strategy is to see whether more labor overlap predicts greater merger gains, and then use heterogeneity tests plus rival/customer/supplier reactions and real effects to determine *which channel* best explains those gains.
- **Main result.** Larger increases in labor-market concentration are associated with higher combined merger wealth effects.
- **Interpretation.** The broader evidence points to *labor efficiency gains* rather than *labor market power* as the main explanation. Merged firms appear to reorganize labor, reduce overlapping employment, increase IT intensity, and become more productive.

7 Conclusion

7.1 Core conclusion

- The paper shows that merger-induced labor market concentration is an important determinant of merger value creation.
- More importantly, it shows that the sign of this value effect is not enough; one needs related-party returns and real outcomes to distinguish labor efficiency from labor power.
- In this sample, the dominant pattern is consistent with labor efficiency gains.

7.2 Economic intuition

- When two firms overlap in labor markets, they are not just buying assets or market share; they are also buying access to an overlapping pool of workers, skills, and tasks.
- That overlap can create value by allowing the merged firm to reassign labor, remove duplication, standardize processes, and substitute technology for redundant labor.
- The key empirical implication is that such mergers should hurt close rivals but help suppliers and customers. That is the exact pattern the paper documents.

7.3 Takeaway

This paper is a strong example of how finance event-study evidence can be combined with labor-market measurement and real-outcome analysis to answer a deeper economic question. The main lesson is not that labor-market concentration is harmless. It is that, in completed U.S. public mergers, the average gains associated with increased labor-market concentration appear to come primarily from efficiency-enhancing reorganization rather than from monopsony-style rent extraction. That conclusion is important for how one interprets merger gains in a world where labor markets matter as much as product markets.